Imperial College London

Department of Electrical & Electronic Engineering

ELEC50001 Circuits & Systems – Mid-term Lab Oral 2026

The mid-term lab oral in the form of a 15-minutes oral examination conducted in person in Level 1 Teaching Lab. This will take place on Monday 2nd or Tuesday 3rd of November 2026. The purpose of this mid-term lab oral is to assess your level of attainment in the intended learning outcomes of **Lab 1 and Lab 2 only**. A schedule for the oral examination is attached. **This is a formal assessment.** If you are unable to attend and wish to swap with someone else, you MUST find a willing person to swap with, and seek my agreement ahead of time via email. The mid-term Lab Oral is worth 15% of the module's marks.

Logbook

You must **bring your logbook** (electronics or paper) with you to the Oral Examination. Although your logbook will NOT be formally "marked", your examiner will nevertheless assess how effective you have kept a log of the experiment and whether you have used your logbook to help you learn through planning and reflection. Your examiner will also be asking questions that can **only** be answered by referring to the logbook.

Format of the Oral – You must turn up to the Level 1 Lab at least 5 minutes before your allocated time. Your Examiner will call you over to their station when it is your turn. You will be asked questions **on Lab 1 and Lab 2 only**. The Examiner may even ask questions not directly linked to what you have done to test your **understanding**. In addition to understanding, examiner will also consider the effort you have put into completing the experiments.

Grading – Attached is the marksheet that your Examiner will be using during the oral examination. The assessment form with feedback will be returned to you via MS Teams. Your will receive a grade for this assessment according to the criteria shown below:

A = Excellent (70%+)	Completed and understood everything with no gap in knowledge
B = Good (60%+)	Completed all and good understanding with minor gaps
C = OK (50%+)	Fair understanding with some gaps, completed most parts
D = Weak (40%+)	Weak understanding with many gaps
E = Fail (below 40%).	Very weak understanding, lack evidence of real effort

We expect the average mark for this assessment to be around 65% for the entire class. To ensure fair assessment, I will perform moderation of marks between different Examiners and apply adjustments if necessary.

Learning Outcomes – Lab 1 and Lab 2

Lab 1 – Amplification

- Characterisation of signal sources with understanding of source impedance and loading effect
- Building of working op-amp circuits in a tidy manner
- Isolating load from source using op-amp using unity gain buffer
- Shifting reference voltage for single power supply operation
- Biasing input and output at middle of supply range
- AC vs DC coupling
- Gain-Bandwidth limitation of op-amp
- Interpretation of op-amp datasheet
- Multi-stage amplification
- Driving of very low output impedance (e.g. speaker) via class D amplifier

Lab 2 – Applications of Op-amps

- Rectifier circuit
- Peak detector and how to choose R C values of the detector
- Voltage comparator with hysteresis and what determines the hysteresis
- Oscillator with positive feedback, and what determines the frequency of oscillation
- Function generator and how it works
- Analogue pulse-width modulator and how it works
- Active filter using Sallen-Key topology
- Butterworth filter design

Peter Cheung

Version 2.1, 24 October 2026.

Mid-term Lab Oral Schedule – 2/3 November 2026

First Name	Surname	Date	Time	Examiner
Raphael	Morris	Monday	09.00 - 09.15	Examiner_1
Zaid	Haris	Monday	09.00 - 09.15	Examiner_2
Ali	Ahmed	Monday	09.00 - 09.15	Examiner_3
Shreya	Rao	Monday	09.15 - 09.30	Examiner_1
Mann	Ng	Monday	09.15 - 09.30	Examiner_2
Yiding Daniel	Zheng	Monday	09.15 - 09.30	Examiner_3
Katherine	Chahbani	Monday	09.30 - 09.45	Examiner_1
Aryaman	Allam	Monday	09.30 - 09.45	Examiner_2
Jiexiang	Lei	Monday	09.30 - 09.45	Examiner_3
Justin	Luo	Monday	09.30 - 09.45	Examiner_4
Shaun	Sundralinghar	Monday	09.30 - 09.45	Examiner_5
Rory	Jones	Monday	09.45 - 10.00	Examiner_1
Pratik	Dhar	Monday	09.45 - 10.00	Examiner_2
Xizhe	Xu	Monday	09.45 - 10.00	Examiner_3
Beiyan	Shi	Monday	09.45 - 10.00	Examiner_4
Shihan	Tan	Monday	09.45 - 10.00	Examiner_5
Zafir	Ahsan	Monday	10.00 - 10.15	Examiner_1
Hywel	Evans	Monday	10.00 - 10.15	Examiner_2
Kerem	Sever	Monday	10.00 - 10.15	Examiner_3
Diya	Palanna	Monday	10.00 - 10.15	Examiner_4
Jayden Wen M	Lim	Monday	10.00 - 10.15	Examiner_5
Devesh	Khemani	Monday	10.15 - 10.30	Examiner_1
Charlie	Roberts	Monday	10.15 - 10.30	Examiner_2
Yaseen	Zoarder	Monday	10.15 - 10.30	Examiner_3
Atchuthan	Senthilkumar	Monday	10.15 - 10.30	Examiner_4
Pin Xie	Loo	Monday	10.15 - 10.30	Examiner_5
Ohm	Laisathit	Monday	10.45 - 11.00	Examiner_1
Han-Yi	Ni	Monday	10.45 - 11.00	Examiner_2
Anthony	Kommatas	Monday	10.45 - 11.00	Examiner_3
Sarthak	Agarwal	Monday	10.45 - 11.00	Examiner_4
Zhengxin	Cai	Monday	10.45 - 11.00	Examiner_5
Bryan	Liow	Monday	11.00 - 11.15	Examiner_1
FONG WAI	WONG	Monday	11.00 - 11.15	Examiner_2
Ananthanaray	Nair	Monday	11.00 - 11.15	Examiner_3
Ayenoya	Alegeh	Monday	11.00 - 11.15	Examiner_4
Seungwon	Na	Monday	11.00 - 11.15	Examiner_5
Tibor	Varga	Monday	11.15 - 11.30	Examiner_1
Zichun	Jin	Monday	11.15 - 11.30	Examiner_2
Xizhe	Xu	Monday	11.15 - 11.30	Examiner_3
Manthan	Dave	Monday	11.15 - 11.30	Examiner_4
Henry	Mak	Monday	11.15 - 11.30	Examiner_5
Tunir	Sarkar	Monday	11.30 - 11.45	Examiner_1
Zhuzhuochen	Ren	Monday	11.30 - 11.45	Examiner_2
Jiexiang	Lei	Monday	11.30 - 11.45	Examiner_3
Syed Ahmed	Iqbal	Monday	11.30 - 11.45	Examiner_4
Yinuo	Huang	Monday	11.30 - 11.45	Examiner_5

First Name	Surname	Date	Time	Examiner
Sami	Benhadj Tahar	Tuesday	09.00 - 09.15	Examiner_1
Hashim	Jastaniah	Tuesday	09.00 - 09.15	Examiner_2
Leal	Bodemeaid	Tuesday	09.00 - 09.15	Examiner_3
Hanting (Justi	Huang	Tuesday	09.00 - 09.15	Examiner_4
Adam	Adamczak	Tuesday	09.00 - 09.15	Examiner_5
Fudail	Fauzy Omar Bashee	Tuesday	09.15 - 09.30	Examiner_1
Seb	Leeming-Price	Tuesday	09.15 - 09.30	Examiner_2
Ian	Tan	Tuesday	09.15 - 09.30	Examiner_3
Zhihan (Andy)	Li	Tuesday	09.15 - 09.30	Examiner_4
Eren	Kusku	Tuesday	09.15 - 09.30	Examiner_5
James	Perillo	Tuesday	09.30 - 09.45	Examiner_1
Suki	Fong	Tuesday	09.30 - 09.45	Examiner_2
Ahmed	Okesola	Tuesday	09.30 - 09.45	Examiner_3
Alexander	Lodge	Tuesday	09.30 - 09.45	Examiner_4
Rafael	Alves	Tuesday	09.30 - 09.45	Examiner_5
Vaishnavi	Rajput	Tuesday	09.45 - 10.00	Examiner_1
Alex	Gray	Tuesday	09.45 - 10.00	Examiner_2
Sahil	Zaki	Tuesday	09.45 - 10.00	Examiner_3
Hosam	Mohamed	Tuesday	09.45 - 10.00	Examiner_4
Alexander	Simpson	Tuesday	09.45 - 10.00	Examiner_5
Oluwadamilo	Omorodion-Banj	Tuesday	10.00 - 10.15	Examiner_1
Waad	Abdulla	Tuesday	10.00 - 10.15	Examiner_2
Yuxuan	Huang	Tuesday	10.00 - 10.15	Examiner_3
Srihari	Karri	Tuesday	10.00 - 10.15	Examiner_4
Morgan	Bleasdale	Tuesday	10.00 - 10.15	Examiner_5
Zoe	Yoong	Tuesday	10.15 - 10.30	Examiner_1
Bruna	Silva	Tuesday	10.15 - 10.30	Examiner_2
Christoper Jie	Koh	Tuesday	10.15 - 10.30	Examiner_3
Finn	Pendlebury	Tuesday	10.15 - 10.30	Examiner_4
Jack	Ward	Tuesday	10.15 - 10.30	Examiner_5
Mishti	Busa	Tuesday	10.45 - 11.00	Examiner_1
Avigat	Gupta	Tuesday	10.45 - 11.00	Examiner_2
Suzanna	MacAlister	Tuesday	10.45 - 11.00	Examiner_3
Mariam	Durrani	Tuesday	10.45 - 11.00	Examiner_4
Fred	Ayres	Tuesday	10.45 - 11.00	Examiner_5
Amina	Mammadzada	Tuesday	11.00 - 11.15	Examiner_1
Yusuf	Yilmaz	Tuesday	11.00 - 11.15	Examiner_2
Aditya	Shivam	Tuesday	11.00 - 11.15	Examiner_3
Oliver	Ho	Tuesday	11.00 - 11.15	Examiner_4
Colin	Qian	Tuesday	11.00 - 11.15	Examiner_5
Fahad	Khan	Tuesday	11.15 - 11.30	Examiner_1
Junchen	Jiang	Tuesday	11.15 - 11.30	Examiner_2
Junxiao	Liang	Tuesday	11.15 - 11.30	Examiner_3
Aziz	Al-Kubaisi	Tuesday	11.15 - 11.30	Examiner_4
Asim	Ozdemir	Tuesday	11.30 - 11.45	Examiner_1
Fong	Si	Tuesday	11.30 - 11.45	Examiner_2
Jiming	Shao	Tuesday	11.30 - 11.45	Examiner_3
Soph	Garrett-Riley	Tuesday	11.30 - 11.45	Examiner_4

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EE2 Circuits and Systems Module
Mid-term Lab Oral

Name of Student: _____

Names of Assessors: _____ Date: Mon / Tue

Performance on the Lab Experiments

1. Logbook Quality and Effectiveness

Highly effective | Effective | OK | Weak | Poor

2. Ability to answer questions from the logbook

Excellent | Good | OK | Poor | Very poor

3. Effort to completing Lab 1 & 2

Fully engaged Strong evidence	Good engagement Good evidence	Acceptable Engagement	Below expected Engagement	V. poor Engagement

Understanding and Learning Outcomes

4. Explanation on theories behind experiments

Excellent | Good | OK | Poor | Very poor

5. Examiner's opinion on candidate's depth of understanding in general

Broad & deep | Good | Average | Less than average | Poor

FEEDBACK TO STUDENT: **Grade:**